

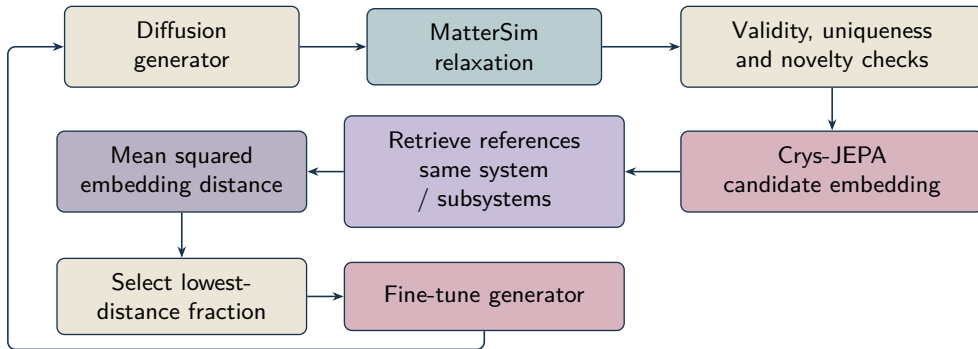
Graph-informed screening and generation of crystals

Enriched reference pools and learned graph context

Screening generated candidates, followed by reference-conditioned generation.

The architectures and improvements discussed here are proposed,
not implemented or experimentally validated results.

Crys-JEPA: existing pipeline



- Crys-JEPA is the crystal encoder; the diffusion generator is a separate model.
- References are whole crystals. The refinement loop does not require new DFT calculations.

Existing method: Liu et al., Crys-JEPA [1]. Relaxation: MatterSim [4].

Crys-JEPA screening score

For a relaxed candidate c , retrieve training crystals whose elements lie within the candidate's chemical system:

$$R(c) = \{r \in \mathcal{D}_{\text{train}} : \text{elements}(r) \subseteq \text{elements}(c)\}.$$

The ranking score is the mean squared distance in Crys-JEPA embedding space:

$$s(c) = \frac{1}{|R(c)|} \sum_{r \in R(c)} \|h_c - h_r\|_2^2.$$

- For an AB_2 candidate, eligible references include A , B , and known A - B phases.
- Each retrieved reference contributes equally. Smaller scores rank first.
- The score measures proximity in an energy-aware representation; it is not an explicit energy-above-hull calculation (this is our understanding).

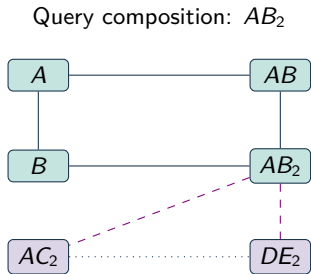
Existing scoring rule: Crys-JEPA [1].

Limitations of the reference pool

- **Chemical-system restriction:** The original pool excludes outside-system crystals that may contain transferable structural or chemical information.
- **Unequal information, equal weights:** References can differ in relevance and redundancy, but the score assigns equal weight to each.
- **Ignores decomposition routes:** An AB_2 candidate can fall to a balanced mix like $AB + B$; averaging individual distances may not directly score this joint competition.
- **Reference-set dependence:** The search pool is fixed, and the score is sensitive to the number of references.

These observations motivate experiments.

Proposed reference graph



Each node is a whole crystal

- Build a broader search pool by considering crystals with structural or chemical similarity. Can be done via a KNN graph in embedding space H .
- For each candidate, learn the references that matter the most, and their ranked order of relevance.

Example. For AB_2 : AB , B rank high (true competitors); AC_2 , DE_2 rank low (context only) $\rightarrow$ keep top- k .

Experiments: does learned reference selection improve screening?

Comparison	Question
Original restricted references vs. enriched search pool	Does access to outside-system analogues help?
Equal-weight references (fixed rule) vs. learned relevance ranking	From the same pool, does learning <i>which</i> references matter beat weighting all of them equally?
Original candidate score; candidate-only predictor; energy predictor + hull	Does the complete method improve candidate screening over simpler alternatives?

Using reference context during generation

The same reference database graph may also inform how a candidate is constructed.

Selection-based refinement

- Score completed candidates.
- Select examples for generator fine-tuning.
- References affect the generator indirectly through training data.

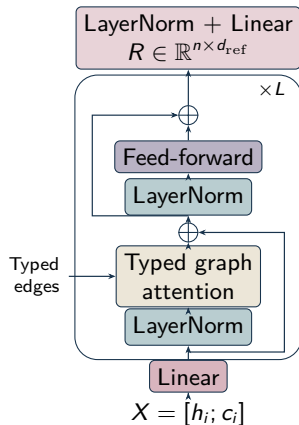
Proposed direct conditioning

- Retrieve references before generation.
- Encode them as contextual tokens.
- Let the denoiser consult these tokens through cross-attention.

The initial scope chooses a target composition before generation. Any improvement over composition alone, such as conditioning on desired properties could also be studied.

Composable Crystals motivates reusable structural conditioning [2]; MatterGen provides diffusion-generation context [3].

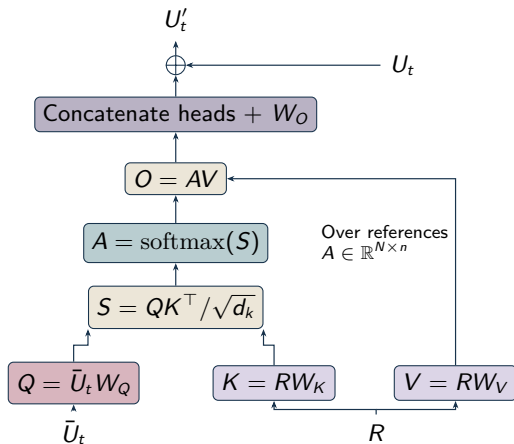
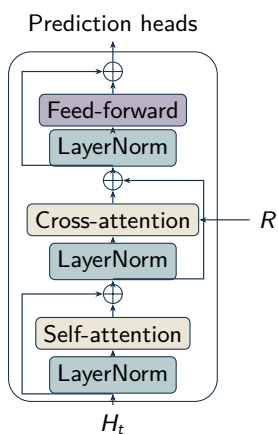
Reference graph encoder



- Input: frozen Crys-JEPA embeddings and composition features for retrieved references.
- Typed graph attention contextualizes each crystal using its neighbours.
- Output: one vector per reference, $R \in \mathbb{R}^{n \times d_{\text{ref}}}$; no global pooling.
- We propose to train cross-attention through the denoising loss to learn which reference information helps reconstruct the candidate.

Representation-learning motivation: I-JEPA predicts target representations rather than pixels [5].

Reference cross-attention



Candidate states supply queries. Reference tokens supply keys and values. Attention selects reference information and adds it to the candidate state.

Why reference information may help

Useful references may help predict the added noise. Under squared-error training, the optimal predictions satisfy

$$f_0 = \mathbb{E}[\epsilon \mid x_t, t, c], \quad f_R = \mathbb{E}[\epsilon \mid x_t, t, c, R], \quad \mathbb{E}\|\epsilon - f_R\|^2 \leq \mathbb{E}\|\epsilon - f_0\|^2.$$

ϵ : added noise; x_t : noisy crystal; t : noise step; c : composition; R : reference information.

- Extra information cannot worsen the optimal prediction.
- Reference crystals give the model examples of known structures. We must test whether these examples improve the generated crystals.
- Retrieve competing phases and similar crystals from other chemistries; test how many to use on validation data.
- Attention learns each reference's influence from prediction errors; this varies by crystal part, attention head and noise step.

Training and inference

Training

1. Choose a crystal and its composition from the **MP-20 training split** [6], following Crys-JEPA [1].
2. Retrieve references using composition; exclude the target and equivalent duplicates.
3. Encode the reference subgraph.
4. Corrupt the training crystal at a sampled diffusion timestep.
5. Optimize the standard denoising loss through the graph encoder, cross-attention and selected denoiser weights.

Currently, inference does not update weights. Selected outputs may support a later refinement round (include with the Crys-JEPA flow)

Inference

1. Specify the requested composition.
2. Retrieve and encode references once.
3. Sample from the diffusion prior and denoise with reference conditioning.
4. Relax the result and apply validity, uniqueness and novelty checks.
5. Retrieve using the completed structure, screen, and select candidates.

Generation evaluation and design choices

Factorial comparison

Generator	Screening
Baseline	Baseline
Baseline	Proposed
Reference-conditioned	Baseline
Reference-conditioned	Proposed

Open design choices

- Fix atom identities/counts, or condition atom-type diffusion on composition.
- Select graph edges, retrieval budget and graph-encoder depth.
- Choose which denoiser weights to fine-tune.
- Consider balanced mixture nodes only if initial screening results motivate them.

Validation of generated crystals

- **Before selection:** measure relaxation success, validity, uniqueness, novelty, diversity and chemical coverage over the full sampled pool.
- **After selection:** independently evaluate the same budget from each generator/screener combination using compatible DFT and hull protocols.
- **Discovery yield:** report independently validated stable or near-hull, unique, novel candidates per generated sample, alongside yield per validated candidate.
- **Reference dependence:** measure similarity to retrieved crystals and control near-duplicates across data splits.
- **Cost and failures:** include generation, retrieval, screening and validation overhead; report unsuccessful calculations.

References

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Questions?